EMS Work-Related Injury Forecast App User Guide

Overview

This guide explains how an EMS manager or operational user can use the EMS Work-Related Injury Forecast app to review historical exposure patterns, generate short-term forecasts, check forecast reliability, and read AI-assisted planning recommendations. The app is built as a Streamlit dashboard that loads weekly exposure data, offers several forecasting methods, and presents the results in a manager-friendly format.

The app is designed around a single weekly time series of work-related exposure counts. It supports forecasting models such as Naive, Moving Average, Exponential Smoothing, Holt, Holt-Winters, and SARIMA, and it can optionally include weather and major-event context when those features are enabled.

What the app does

The main purpose of the app is to help EMS leaders move from reactive reporting to proactive planning. Instead of waiting until a cluster of work-related exposure events has already happened, the app allows users to estimate near-term exposure levels and view those estimates together with backtest accuracy metrics.

The app also includes an AI Planning Assistant, called PAi, that can translate forecast outputs into practical recommendations about staffing, safety messaging, training, and PPE planning. The AI output is based on recent history, the selected model, the forecast horizon, and the model's recent forecast error.

Before you begin

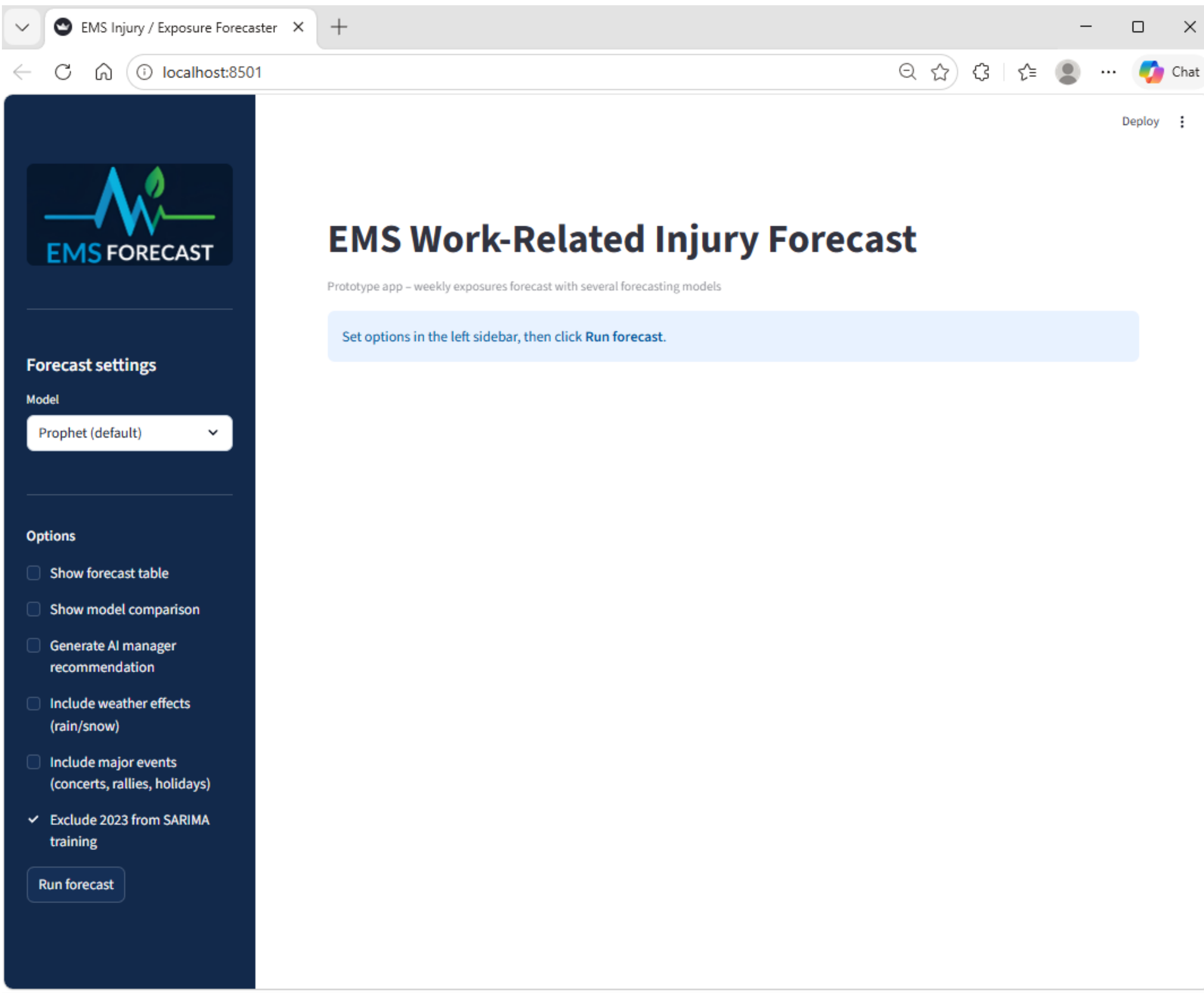
Before using the app, make sure the application is running and that the weekly exposure data file is available in the expected project path. The app reads a CSV-based exposure dataset, parses timestamps, aggregates daily counts, and resamples those counts into Monday-based weekly exposure totals.

The weather and event options are optional features. Weather is pulled from Open-Meteo when overlapping dates are available, and event counts are fetched from the Ticketmaster Discovery API for the Greater Vancouver area.

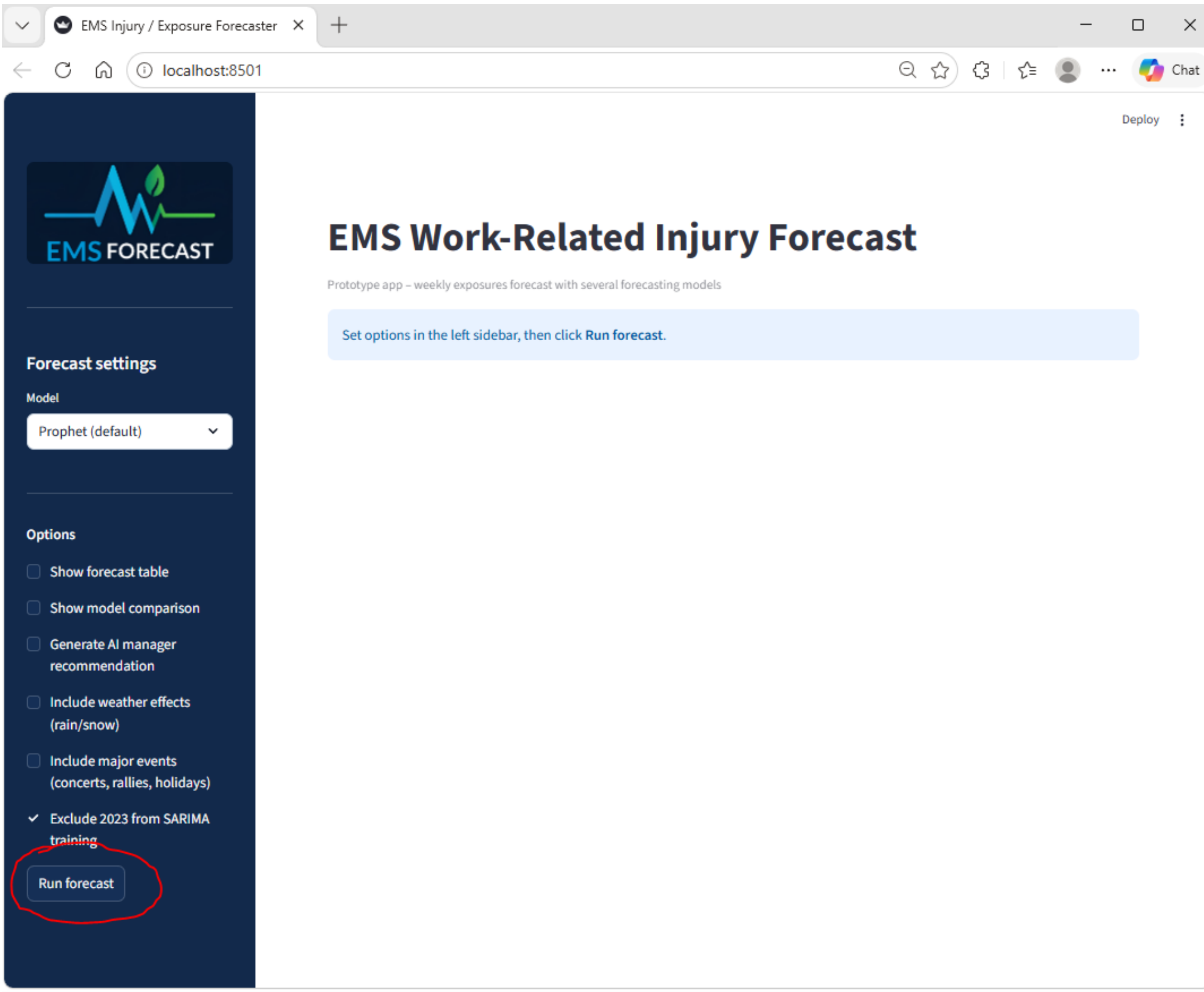
Step-by-step instructions

1. Open the app

Launch the Streamlit application and wait for the dashboard to load. The title shown at the top of the app is “EMS Work-Related Injury Forecast,” and the subtitle describes it as a prototype app for weekly exposure forecasting with several forecasting models.

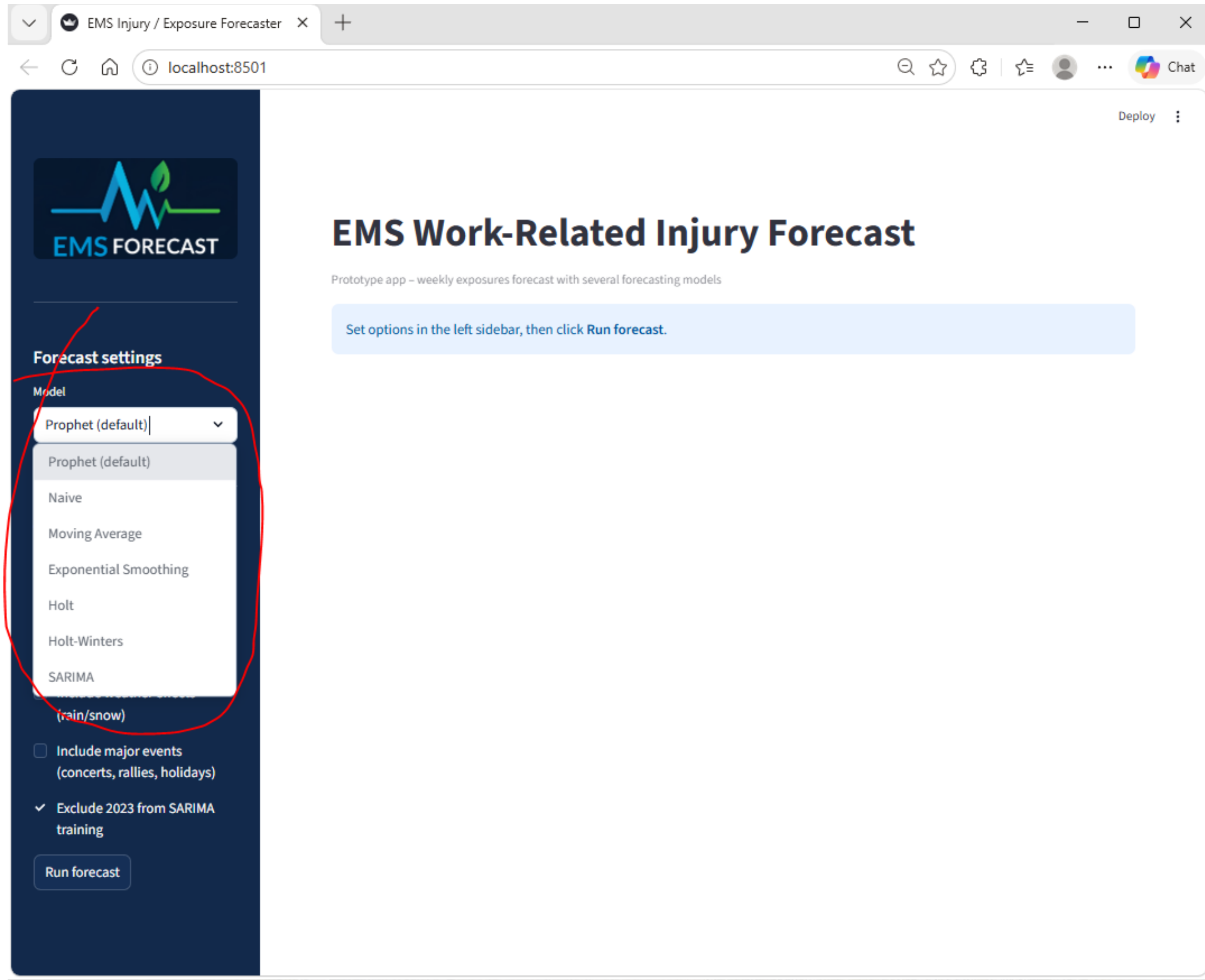


When the app first opens, it may show an information message asking the user to set options in the left sidebar and click **Run forecast**. This is normal behavior and means no forecast has been executed yet in the current session.

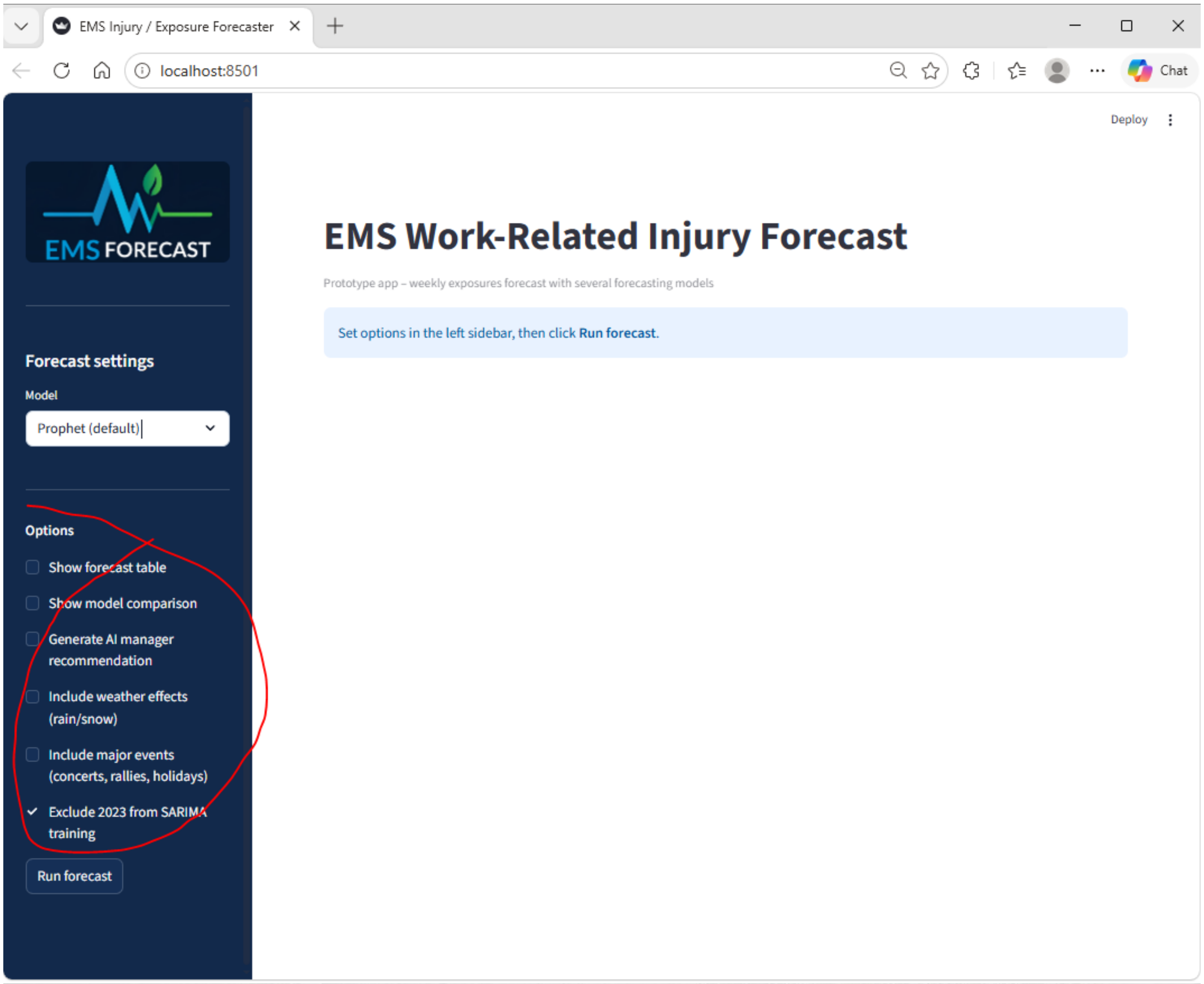


2. Review the sidebar settings

The left sidebar contains the main forecast setup area. Under **Forecast settings**, the user can choose a forecasting model from options including “Prophet (default),” “Naive,” “Moving Average,” “Exponential Smoothing,” “Holt,” “Holt-Winters,” and “SARIMA.”



Below the model selector, the **Options** section includes several checkboxes that let the user control what appears in the results. These include **Show forecast table**, **Show model comparison**, **Generate AI manager recommendation**, **Include weather effects (rain/snow)**, **Include major events (concerts, rallies, holidays)**, and **Exclude 2023 from SARIMA training**.



3. Choose the forecast model

Start with a simple model if the goal is a quick operational estimate. The Naive, Moving Average, Exponential Smoothing, and Holt models are the easiest to interpret and are well suited for users who want transparent short-term forecasts.

Use SARIMA if the goal is to capture more time-series structure and the dataset is large enough after filtering. The app checks that enough clean history is available before fitting SARIMA and stops with an error message if too little usable history remains.

4. Select optional features

Turn on **Show forecast table** if a week-by-week numeric output is needed. This option is useful when managers want to read exact point forecasts and prediction bounds rather than relying only on the chart.

Turn on **Generate AI manager recommendation** when a plain-language operational summary is needed. This feature sends a structured forecast summary and backtest metrics to the AI recommendation function so the app can generate a short recommendation for non-technical EMS managers.

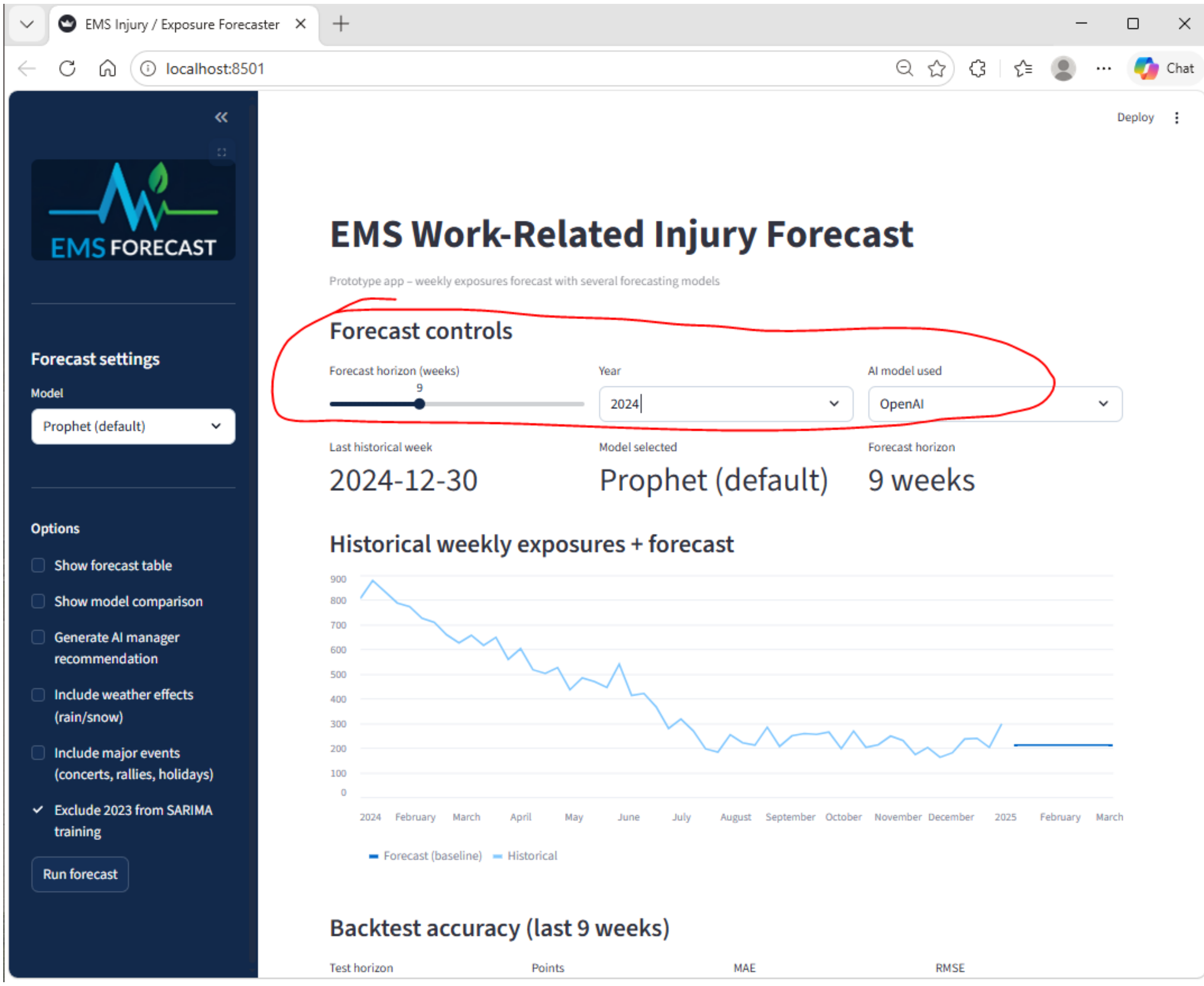
Turn on **Include weather effects (rain/snow)** only when weather context is relevant. When enabled, the app trims the history to weeks where weather data are available, retrieves daily precipitation, snowfall, and temperature data from Open-Meteo, and aggregates them to weekly values.

Turn on **Include major events (concerts, rallies, holidays)** if the user wants to test whether local events might be related to operational risk. When enabled, the app calls the Ticketmaster Discovery API, counts the number of events per week, and can display the retrieved event table in the dashboard.

5. Run the forecast

After choosing a model and any optional settings, click the **Run forecast** button in the sidebar. This activates the main forecast view and loads the control panel in the main page area.

Once the forecast is running, the app displays a **Forecast controls** section in the main panel. This includes a **Forecast horizon (weeks)** slider from 1 to 24 weeks, a **Year** filter, and an **AI model used** selector with options for Gemini, OpenAI, or None.



6. Set the forecast horizon

Use the **Forecast horizon (weeks)** slider to choose how many future weeks to predict. A shorter horizon is generally easier to interpret and tends to be more useful for immediate staffing and safety planning, while a longer horizon can be used for broader operational discussion.

The app builds future weeks as Monday-based weekly timestamps. Every model returns the same core forecast structure: `week_start`, `yhat`, `yhat_lower`, and `yhat_upper`.

7. Review the historical series and forecast output

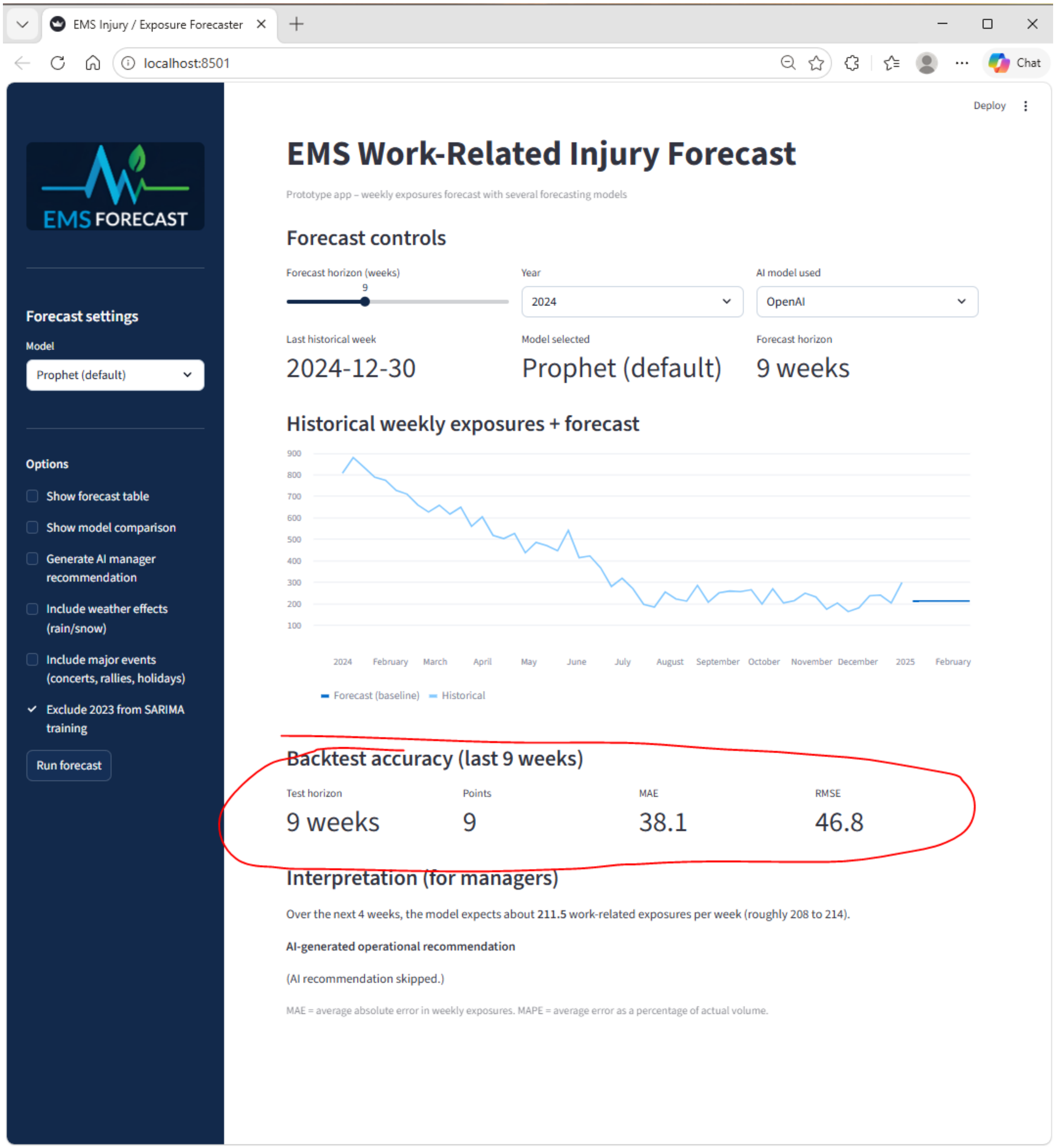
After the forecast runs, review the main results area. The application overlays the forecast on top of the historical weekly exposure series so that users can visually compare recent observations with the projected near-term trend.

If the forecast table option is enabled, the app also shows a week-by-week table containing the forecast date, point forecast, and upper and lower bounds. This is useful for documenting expected volume and discussing operational scenarios in meetings.

8. Interpret the backtest metrics

The app includes a backtesting routine that holds out recent weeks, fits the selected model on earlier data, and evaluates performance on the held-out period. It reports standard metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Use **MAE** to understand the typical size of the forecast miss in units of exposures per week. Use **RMSE** to see whether some weeks had larger misses than average, since RMSE increases more when large errors occur.

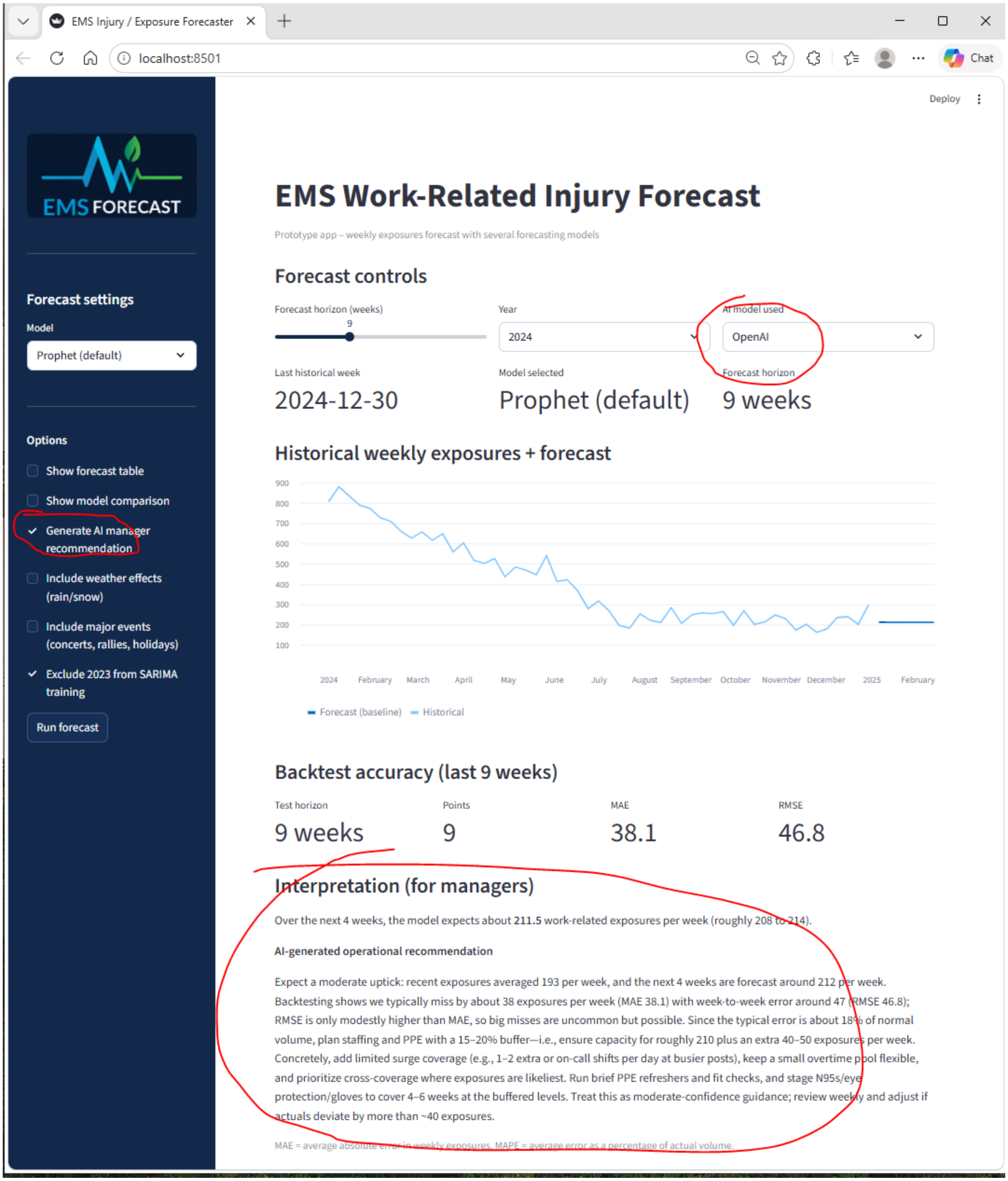


The app also computes MAE as a percentage of the average weekly exposure level. This helps managers interpret whether the model's average error is small enough to support short-term planning decisions.

9. Read the AI Planning Assistant output

If the AI recommendation option is enabled, the app generates a short narrative for managers. The recommendation is built from a structured summary that includes the recent 8-week average, the next 4-week forecast average, the selected model, the forecast horizon, and the backtest metrics.

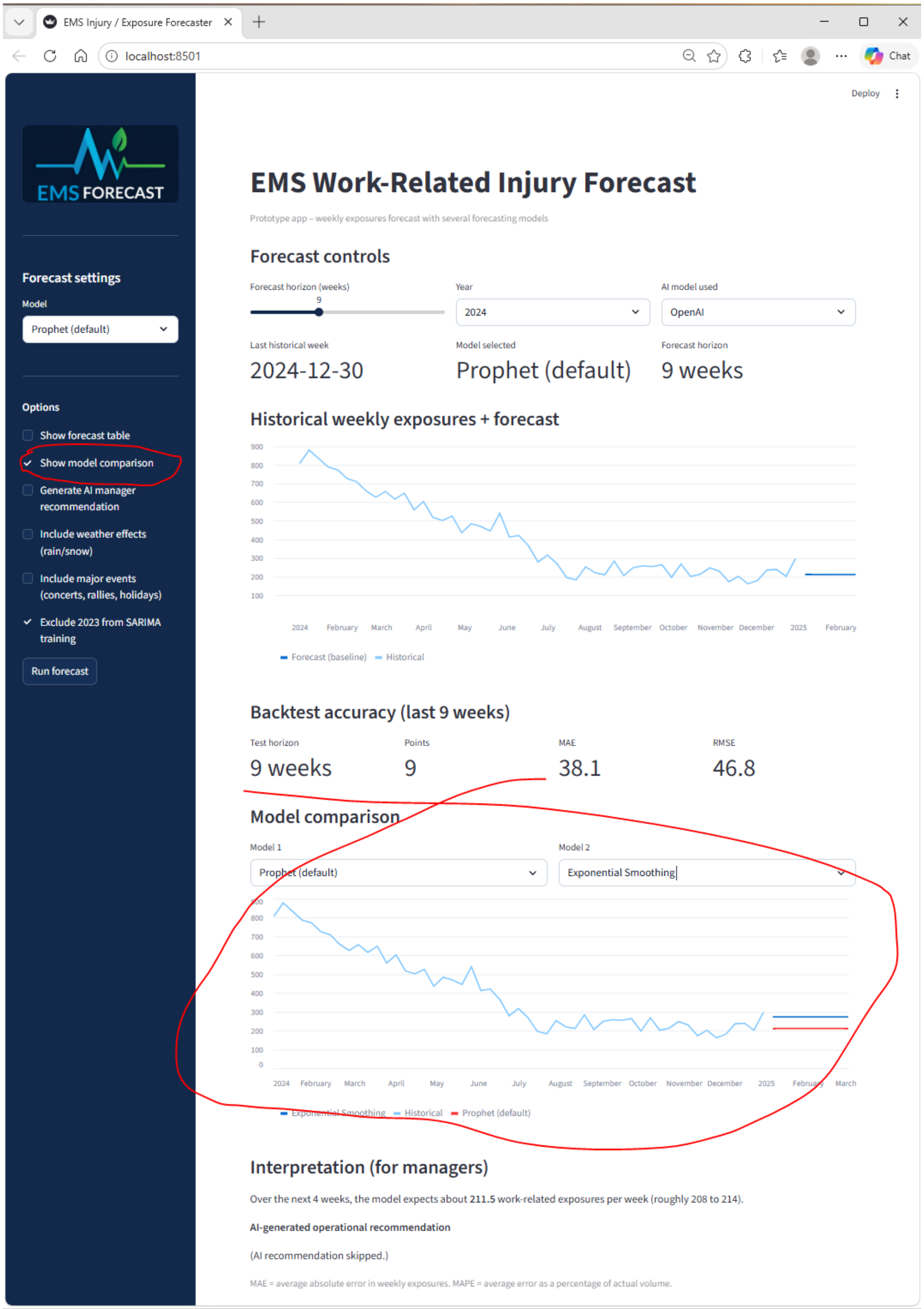
The AI recommendation is intended to be practical and non-technical. It is designed to explain average forecast error in plain language, note whether larger misses are possible, and suggest actions related to staffing, training, and PPE planning.



10. Compare models and rerun

Users can rerun the forecast with a different model at any time. This is helpful when comparing a simple baseline such as Naive or Moving Average against a more structured approach such as Holt or SARIMA.

If the **Show model comparison** option is enabled, use that section to review differences in output across models. In practice, a model should not be selected only because its curve looks better visually; it should also be judged by its recent backtest accuracy.



How to use the outputs in practice

A practical way to use the app is to begin with a short forecast horizon, such as 1 to 4 weeks, and compare two or three interpretable models. Then review the recent error metrics and decide whether the forecast is reliable enough to support actions such as targeted safety reminders, staffing adjustments, or extra PPE emphasis.

The best operational use of the app is not to treat the forecast as a guaranteed prediction. Instead, it should be treated as a planning signal that helps managers prepare for possible higher-risk periods while keeping forecast uncertainty visible.

Troubleshooting tips

If the app says there is not enough history for a given model, switch to a simpler model such as Holt or Naive. For Holt-Winters, the code checks for at least two seasonal cycles of weekly data before allowing the model to run.

If weather data are unavailable, the app will display a message and continue without weather effects. If Ticketmaster returns no events for the selected period, the app will display an informational message rather than failing.

If SARIMA fails after filters are applied, try including more years or disabling the option that excludes 2023 from SARIMA training. The code specifically checks for a minimum amount of clean data after filtering and stops when the remaining series is too short.

Suggested workflow for managers

1. Open the app and select a simple model first.
2. Choose a short forecast horizon, such as 1 to 4 weeks.
3. Run the forecast and inspect the chart.
4. Review MAE, RMSE, and MAPE before making decisions.
5. Turn on the AI recommendation for a plain-language planning summary.
6. Compare models only after checking whether the historical data window is adequate.
7. Use the forecast as a planning aid, not as a certainty.